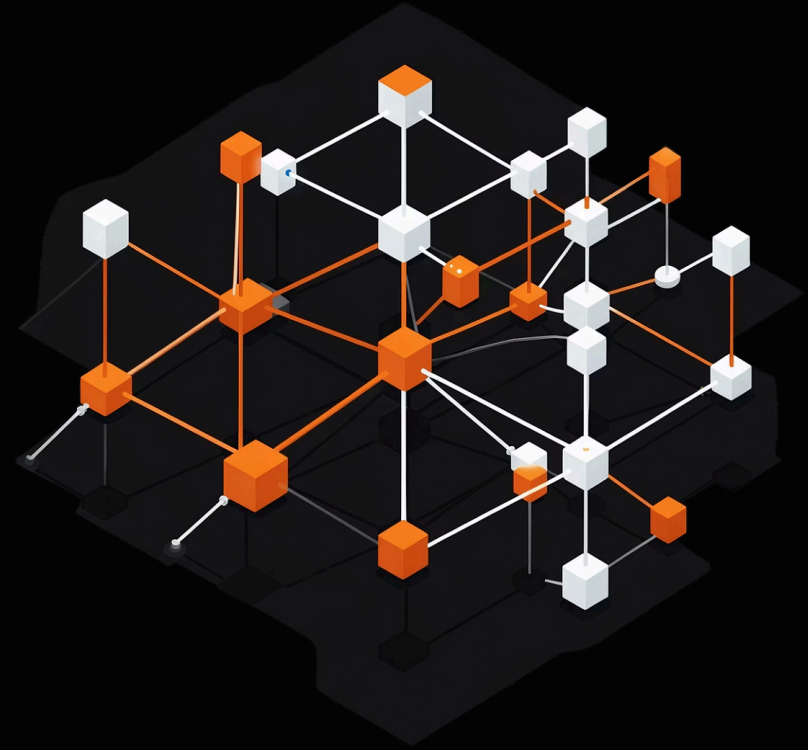


# Graph Theory Basics

An introduction to vertices, edges, and the mathematics of connections.





# What is a Graph?

## Vertices & Edges

A graph is a collection of **vertices** (nodes) connected by **edges** (links) — the two fundamental building blocks.

## Real-World Models

Graphs appear everywhere: social networks, road maps, computer networks, and friendship connections.

## Undirected Graphs

Edges have **no direction** — a connection between A and B means travel is possible both ways.



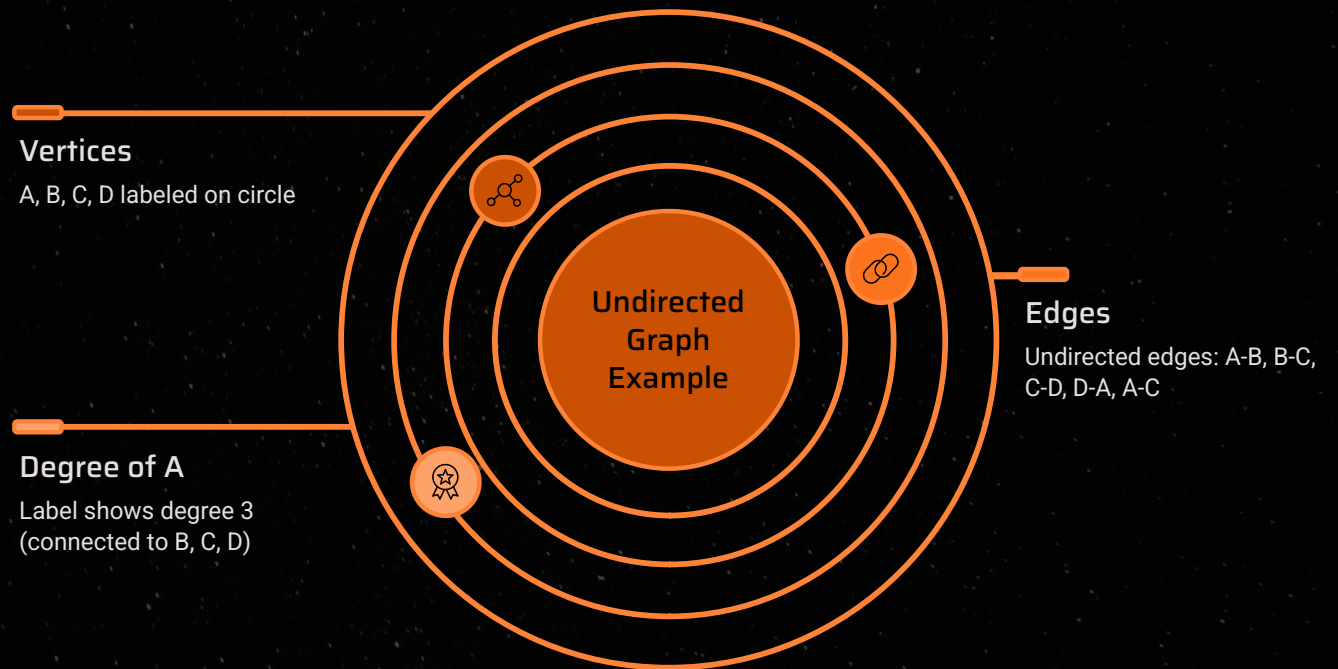
# A Simple Example

## The Graph

Consider a graph with **4 vertices**: A, B, C, and D, connected by the following edges:

- A–B
- B–C
- C–D
- D–A
- A–C

❏ Vertex **A** has degree **3** — it connects to B, D, and C.



# Adjacency Matrix

## What Is It?

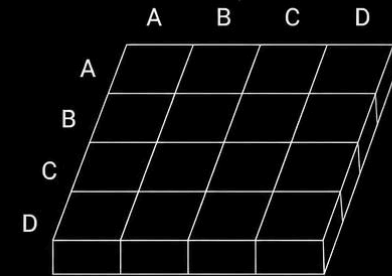
An **adjacency matrix** is a table that encodes which vertices are connected in a graph.

- Rows and columns represent each vertex
- Entry is **1** if an edge exists between the two vertices
- Entry is **0** if there is no edge
- For undirected graphs, the matrix is always **symmetric**

Adjacency matrix concept:

|   | A | B | C | D |
|---|---|---|---|---|
| A | 0 | 1 | 1 | 0 |
| B | 1 | 0 | 1 | 1 |
| C | 1 | 1 | 0 | 1 |
| D | 0 | 1 | 1 | 0 |

A symmetric 4x4 grid labeled A, B, C, D on both axes.



Cells contain 0 or 1.

|   | A | B | C | D |
|---|---|---|---|---|
| A | 0 | 1 | 1 | 0 |
| B | 1 | 0 | 1 | 1 |
| C | 1 | 1 | 0 | 1 |
| D | 0 | 1 | 1 | 0 |

Highlight the symmetry — top-right mirrors bottom-left.

|   | A | B | C | D |
|---|---|---|---|---|
| A | 0 | 1 | 1 | 0 |
| B | 1 | 0 | 1 | 1 |
| C | 1 | 1 | 0 | 1 |
| D | 0 | 1 | 1 | 0 |

# Reading an Adjacency Matrix

For our 4-vertex graph (A, B, C, D) with edges A–B, B–C, C–D, D–A, A–C:

|   | A | B | C | D |
|---|---|---|---|---|
| A | 0 | 1 | 1 | 1 |
| B | 1 | 0 | 1 | 0 |
| C | 1 | 1 | 0 | 1 |
| D | 1 | 0 | 1 | 0 |

Row A, Column B = **1** means A and B are connected. Because the graph is undirected, Row B, Column A is also **1**. The diagonal is always **0** — no vertex connects to itself.

# Paths in Graphs



## Path

A sequence of vertices where each consecutive pair is connected by an edge — the route you travel through a graph.



## Simple Path

A path in which **no vertex appears more than once** — no revisiting allowed along the way.

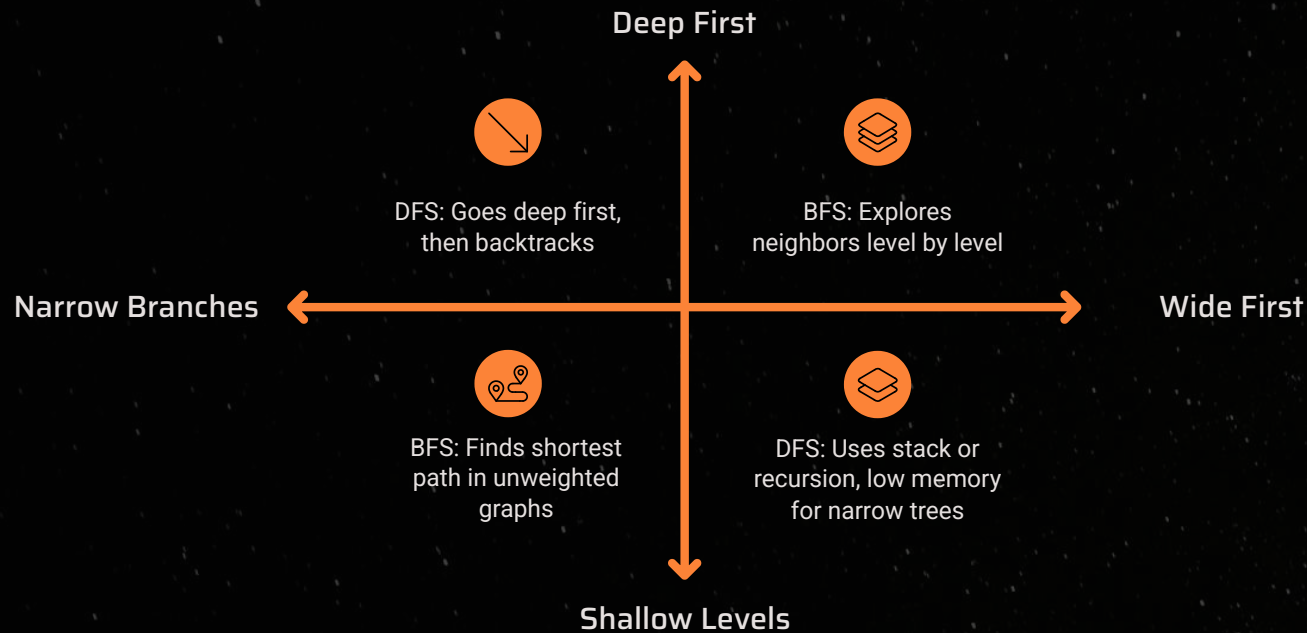


## Cycle

A path that **starts and ends at the same vertex**, forming a closed loop through the graph.



# Finding Paths



## Two Classic Algorithms

Both algorithms answer the core question:  
**does a path exist between two vertices?**

1

### Depth-First Search (DFS)

Explore as far as possible along each branch before backtracking. Like solving a maze by always going forward until you hit a dead end.

2

### Breadth-First Search (BFS)

Explore all immediate neighbors before moving deeper. Radiates outward level by level from the starting vertex.

# Key Takeaways



## Graphs Model Relationships

Vertices and edges capture connections between real-world entities — from friends to cities to web pages.



## Adjacency Matrices Encode Structure

A symmetric table of 0s and 1s gives us a precise mathematical representation of any undirected graph.



## DFS and BFS Explore Graphs

These two fundamental algorithms let us systematically find paths and understand the connectivity of any graph.

📌 Master these building blocks and you'll have the foundation for advanced topics like shortest paths, spanning trees, and network flow.